Contents

Why the de Sitter vacuum cannot fix the MOND acceleration scale: the required moment is the pole of ζ

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Version 1 (2026-08-08).

The result in one line

For a modified-inertia worldline action whose memory kernel is the de Sitter vacuum autocorrelation, the acceleration scale is fixed by that kernel's **first** moment. All the moments have one closed form, and

$$\boxed{M_p \;=\; \int_0^\infty ds\; s^p\,\frac{a^2}{\sinh^2(as)} \;=\; 2^{\,1-p}\,\Gamma(p{+}1)\,\zeta(p)\,a^{1-p}\;, \qquad a = \tfrac{H}{2} = \frac{\pi}{\beta}}$$

so the moment the theory needs, $p = 1$, is **exactly the pole of ζ** — the only pole ζ has. The divergence is not a technical nuisance to be renormalised away and then quietly used: it is the same fact that forbids the required $\sqrt{\pi}$, because a half-integer power of π enters M_p only through $\Gamma(3/2) = \sqrt{\pi}/2$ at *half-integer* p, and the rapidity gap $\theta = (s/c)|a|$ is *linear* in s, which forces $p = 1$ and forbids every half-integer.

One integer therefore carries the whole obstruction. The consequence is positive as well as negative: $M_1 = c/a_0$ is a **renormalisation condition**, not a computable number, and a_0 is the subtraction point of the theory rather than one of its outputs.

What this is not. It is not a derivation of the MOND coefficient, and it is not a step toward one — it is a proof that one particular and much-travelled route cannot produce it. Two further routes are closed in §7 by the same π-parity mechanism. §8 states plainly what remains open.

1. What is being asked

Milgrom's vacuum reading of MOND [1] puts a constant-acceleration observer in de Sitter space, identifies a_0 with $\sqrt{\Lambda/3}$, and forms the temperature excess $\Delta T = T(a) - T(0)$ with $T \propto \sqrt{a^2 + a_0^2}$ — a structure that reproduces both MOND limits. The interpolating function that follows, $\nu = \sqrt{1 + 1/y}$, is Eq. (9) of that paper; it is not a variant of it, and nothing about the kernel is new here. Milgrom's companion result [2] is equally load-bearing for what follows: modified-inertia theories are *generically time-nonlocal*. The memory-kernel framing is his.

In a worldline realisation of that picture [3] the nonlocality is explicit,

$$S \;=\; -mc^2 \int \Big[\mu(\Theta)\, d\tau + (1 - \mu(\Theta))dt\Big], \qquad \Theta(\tau) \;=\; \int_0^\infty ds\, K(s)\, \cosh^{-1}\Big(-\tfrac{u(\tau)\cdot u(\tau - s)}{c^2}\Big),$$

and the acceleration scale enters through exactly one number, the kernel's first moment $M_1 = \int_0^\infty s\, K(s)\, ds$. With the memory force of general orbits included, $a_0 = \frac{2}{3} c/M_1$, and the coefficient question becomes a proved equivalence:

$$\kappa = \tfrac{1}{2} \quad \Longleftrightarrow \quad M_1 = \tfrac{4}{3}\,(G\rho_\Lambda)^{-1/2} = \tfrac{4}{3}\, t_\Lambda, \qquad t_\Lambda = 50.74 \text{ Gyr},$$

where $a_0 = \kappa\, c\sqrt{G\rho_\Lambda}$ and $\kappa = \frac{1}{2}$ gives the observed $9.36 \times 10^{-11}\,\mathrm{m\,s^{-2}}$. Equivalently, in units of the thermal time $\beta = 2\pi/H_\Lambda$,

$$\frac{M_1}{\beta} \;=\; \sqrt{\frac{32}{27\pi}} \;=\; 0.61421182,$$

a footing-invariant target: it is numerically identical under the canonical footing (ρ_{DE} with cH_Λ, $a_0 = 9.3619 \times 10^{-11}$, $t_\Lambda = 50.74$ Gyr) and under the alternative footing (a factor 1.2048 larger, $a_0 = 1.1279 \times 10^{-10}$, $t_\Lambda = 42.11$ Gyr), because both M_1 and β scale the same way. Nothing below depends on that choice.

The question of this paper is therefore sharp and self-contained. **The kernel K is a vacuum autocorrelation. Compute its first moment. Does it equal $\frac{4}{3}t_\Lambda$?**

2. The correlator is exactly thermal

Theorem 1. *Along a geodesic in dS_4, the conformally coupled massless scalar two-point function restricted to the worldline is*

$$k(s) \;=\; \frac{a^2}{\sinh^2(as)}, \qquad a = \frac{H}{2} = \frac{\pi}{\beta}, \qquad \beta = \frac{2\pi}{H},$$

and this function is thermal at $T = H/2\pi$ in the strict KMS sense.

Three independent verifications are carried in the accompanying script, none of them by citation:

1. **KMS periodicity.** $k(s+i\pi/a)-k(s)=0$ identically, i.e. periodicity in imaginary time with period $\beta=\pi/a=2\pi/H$. This *is* the KMS condition; the temperature is not inserted, it is read off.
2. **Coincidence with the flat thermal correlator.** $(\pi/\beta)^2/\sinh^2(\pi s/\beta)$ at $\beta=\pi/a$ is the same function, symbol for symbol. De Sitter along a geodesic *is* a heat bath, not merely thermal-like.
3. **Short-distance structure.** $k(s)=1/s^2-H^2/12+O(s^2)$. The singularity is the *flat* $1/s^2$ with no H in it — so the Hadamard subtraction is exactly $1/s^2$ — and the first correction is pure curvature. A control confirms this detects curvature rather than restating a trivial identity: the flat correlator's subtracted limit is exactly zero, de Sitter's is $-H^2/12$.

The large-s behaviour is $k\to 4a^2e^{-Hs}$, decay at the thermal rate. This is all classical [4,5,6]; it is stated with verification because everything after it is arithmetic on this function.

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3. All the moments in one closed form

Theorem 2. *With $1/\sinh^2 x=4\sum_{n\geq 1} n\,e^{-2nx}$ (verified numerically to $<10^{-30}$),*

$$M_p\;=\;\int_0^\infty ds\; s^p\,k(s)\;=\;2^{\,1-p}\,\Gamma(p{+}1)\,\zeta(p)\,a^{1-p}.$$

The underlying Mellin transform $\int_0^\infty x^{z-1}\sinh^{-2}x\,dx\;=\;2^{2-z}\Gamma(z)\zeta(z{-}1)$ is a tabulated classical integral; the content here is what it says about $p=1$.

Verification matters because p ranges over both convergent and continued regimes:

p	closed form	quadrature	relative error	route
1/2	−1.531324582669376	−1.531324582669376	9.2×10^{-23}	subtracted
3/2	2.934997656735202	2.934997656735202	9.5×10^{-23}	bare
2	2.349905809783181	2.349905809783181	0	bare
3	3.679766030080391	3.679766030080391	1.3×10^{-41}	bare
4	9.466383968318993	9.466383968318993	1.9×10^{-41}	bare

For $p>1$ the bare integral converges. For $p<1$ it power-diverges at $s\to0$ and the Hadamard-subtracted integral is the right object — and the two agree, which is what licenses using the continuation at all: at $p=0$ the closed form gives **exactly** $-a$ (because $\zeta(0)=-1/2$), and the physically subtracted quadrature reproduces $-a$ to 10^{-25}. The analytic continuation and the physical subtraction are the same map.

Corollary 2.1 (the pole). ζ has exactly one pole, at $p=1$, and

$$(p-1)\,M_p\;\longrightarrow\;1\qquad(p\to1),$$

verified as 1.0067, 1.00007, 1.0000007, 1.0000000066 at $p-1=10^{-2},10^{-4},10^{-6},10^{-8}$: **a simple pole with residue exactly 1, scheme-independently.** Every other integer moment

$p = 0, 2, 3, 4, 5$ is finite, so the divergence is specific, not generic. In closed form the divergence is logarithmic with unit coefficient,

$$\int_{\delta}^{\infty} s\,k(s)\,ds \;=\; 1 - \ln 2 + \ln\frac{1}{a\delta},$$

confirmed against quadrature to 10^{-12} and, independently, by the numerical growth rate: the integral rises exactly linearly in $\ln(1/\delta)$ with slope 1. A further control: `mpmath` itself refuses $\zeta(1)$, so the pole is asserted by the library rather than by the author.

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4. And $p = 1$ is forced

Theorem 3. *The rapidity gap is linear in the lag, so the action pairs K against s^1 and against no other power.*

$-u \cdot u'/c^2 = \cosh(w - w')$ depends only on the rapidity difference; for hyperbolic motion $w(\tau) = |a|\tau/c$ exactly, so

$$\theta(\tau, \tau - s) \;=\; \frac{s}{c}\,|a|, \qquad \frac{\partial\theta}{\partial s} = \frac{|a|}{c}, \qquad \frac{\partial^2\theta}{\partial s^2} = 0,$$

and for a general worldline the midpoint rule gives $\theta = (s/c)|a(\tau - s/2)| + O(s^3)$ [3]. Hence $\Theta = (|a|/c)\int sK(s)\,ds$ and the only moment that appears is M_1.

Combining Theorem 3 with Corollary 2.1: the one moment the framework requires is the one moment the correlator does not have.

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5. The same integer forbids the $\sqrt{\pi}$

This is the part that turns a divergence into a structural statement, because it shows the divergence and the missing irrationality are not two problems.

Theorem 4 (π-parity of the moments). *At integer p, M_p has integer π-weight; a half-integer π-weight occurs only at half-integer p.*

p	M_p/a^{1-p}	π-weight
0	-1	0
2	$\pi^2/6$	2
3	$\frac{3}{2}\zeta(3)$	0
4	$\pi^4/30$	4
1/2	$\frac{\sqrt{2}}{2}\sqrt{\pi}\,\zeta(1/2)$	1/2
3/2	$\frac{3\sqrt{2}}{8}\sqrt{\pi}\,\zeta(3/2)$	1/2
5/2	$\frac{15\sqrt{2}}{32}\sqrt{\pi}\,\zeta(5/2)$	1/2

The mechanism is transparent: $\Gamma(p+1)$ is rational at integer p and carries $\sqrt{\pi}$ at half-integer p, while $\zeta(\text{even})$ is a rational multiple of an even power of π and $\zeta(\text{odd})$ is π-free.

The target is

$$\xi \;\equiv\; M_1 H_\Lambda \;=\; \frac{8\sqrt{6}\sqrt{\pi}}{9} \;=\; \frac{2^{7/2}\sqrt{\pi}}{3^{3/2}} \;=\; 3.85920669, \qquad \xi^2 = \frac{2^7\pi}{3^3} = 2\pi\left(\tfrac{4}{3}\right)^3,$$

of π-weight $+1/2$ exactly. **So the correlator could supply the target's $\sqrt{\pi}$ only at $p = 1/2$ or $p = 3/2$, which §4 forbids.** The divergence and the missing $\sqrt{\pi}$ are one fact: $p = 1$.

This is a specific exclusion, not a blanket one. Four prespecified decoy targets — $2/3$, $\pi/6$, $\zeta(3)$, $\pi^2/16$ — all have integer π-weight and would therefore be reachable by an integer-p moment ratio. The machinery admits them and rejects the actual target. A test that rejected everything would prove nothing.

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6. The memory time: a two-pronged no-go

One might hope to sidestep §3 by asking not for M_1 but for the kernel's *correlation time* $\tau_c = M_1/M_0$, which is free of the coupling. Both available schemes fail, in opposite directions.

Prong 1 — unsubtracted: no memory at all. M_0 diverges as $1/\delta$ (a *power*) while M_1 diverges only logarithmically, so

δ	M_0	M_1	τ_c
10^{-3}	999.3	7.571	7.58×10^{-3}
10^{-6}	10^6	14.479	1.45×10^{-5}
10^{-9}	10^9	21.387	2.14×10^{-8}
10^{-12}	10^{12}	28.295	2.83×10^{-11}

$\tau_c \to 0$. **The bare de Sitter correlator has zero memory** — it is UV-dominated, and a UV-dominated kernel cannot supply a cosmological time.

Prong 2 — Hadamard-subtracted: the infrared takes over. Now $M_0 = -a$ *exactly* (finite, and it does carry the thermal scale, which is the encouraging part), but M_1 diverges in the **infrared** as $-\ln(aS)$: the combination $M_1^H + \ln(aS)$ converges to 0.30685281944 as the IR cutoff runs over four decades. There is no scheme in which both moments are finite.

One exact curiosity, reported because it is exact and not because it predicts anything. Cutting the logarithm at the horizon itself, $\delta = 1/H$, gives the dimensionless first moment **exactly 1** — the $\ln 2$ from $a = H/2$ cancels the $-\ln 2$ of the closed form. It is a clean coincidence and it fixes no dimensionful number.

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7. What the failure means, and two companion no-goes

7.1 a_0 is a renormalisation condition

k has dimension $1/\text{time}^2$ while the action's K has $1/\text{time}$, so $K = C\,k$ with $[C] = \text{time}$, and $M_1 = C \times (\text{dimensionless moment})$. Both C and the subtraction point are free. **This is precisely**

the structure the worldline realisation already has — $K = (N/\lambda)e^{-s/\lambda}$ with only the *product* $N\lambda = M_1$ fixed — and it explains that structure rather than merely restating it: a free-field correlator cannot deliver M_1, because the object is a logarithmically divergent moment, and the finite part of a divergent moment is a renormalisation condition. a_0 **is the subtraction point of the theory.**

The mismatch is robust in the way logarithms are robust. Taking the subtraction at a UV scale:

subtraction δ	implied $\hat{M}_1$	ratio to the target 0.61421
Planck time	141.48	73.3
nuclear, 10^{-23} s	94.81	49.1
atomic, 10^{-16} s	78.70	40.8
one second	41.85	21.7
required	—	6.92 Gyr

Forty-three decades of cutoff move the answer by a factor of 3.4 and never approach the target, and the subtraction that *would* reproduce a_0 sits at 6.92 Gyr — a cosmological time, not a short-distance scale. So this is not a UV renormalisation at all.

Stated against my own framing: §7.1 is not a kill. Because C is free, a Planck-scale subtraction gives $\tau_c = 7.6 \times 10^{-42}$ s, which *passes* the ephemeris bound $\lambda \leq 39$ yr comfortably, at kernel weight $N = 2.8 \times 10^{59}$ — and N is a free coupling. Nothing in this table excludes anything. Nothing in it predicts anything either. **The kill is Corollary 2.1 with Theorems 3 and 4, not the arithmetic of cutoffs.**

7.2 Companion no-go: the mode sum over the $SO(1,3)$ generators

The target's algebraic part is $(8/9)\sqrt{6}$, and $\sqrt{6} = \sqrt{\dim \mathfrak{so}(1,3)}$ — verified from the D_2 root data (rank 2 plus 4 roots) and the $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ split, with $D = 4$ the **only** integer satisfying $D(D-1)/2 = 6$. That is suggestive. It does not survive.

- **The $\sqrt{\pi}$ cannot be group-theoretic.** Every sphere volume $\mathrm{Vol}(S^{n-1}) = 2\pi^{n/2}/\Gamma(n/2)$ has *integer* π-weight for $n = 2, \ldots, 10$, because Γ at half-integer argument returns the compensating $\sqrt{\pi}$; and $\mathrm{Vol}(SU(2)) = 2\pi^2$, $\mathrm{Vol}(SO(4)) = 2\pi^4$ are π-even. So the half-integer weight has exactly one address, the odd-dimensional momentum measure $(4\pi)^{-3/2}$, and no mode sum can reach it.
- **And** $8/9$ **is not selected.** Closing the canonical invariants of $\mathfrak{so}(1,3)$ (dimension, rank, dual Coxeter number, adjoint Casimir, Weyl group order, positive-root count, and the $\mathfrak{su}(2)$ factor's dimension and Casimir) under ratios of products of at most two gives a menu of **33** distinct rationals, of which **30** are equally "nice-looking" and which contains all four prespecified decoys $9/8$, $4/3$, $3/4$, $16/9$. A prespecified entry carries $p = 0.030$. *This is more favourable than the "hundreds" I asserted before running it, and the correction is recorded rather than absorbed.*
- **Decisively, the Casimir is convention-dependent.** C_2(adjoint) is 4 in the highest-root normalisation and 2 in the spin normalisation — a factor of 2, which is exactly the size of the quantity such a derivation would be asked to explain.

The route dies, and the $\sqrt{6}$ is downgraded to a coincidence. What survives is only the reduction $\kappa = Z/(3\xi)$ with $\xi = 2Z/3$.

7.3 Companion no-go, and a withdrawal against interest

A previous note of mine observed that $\kappa = \frac{2}{3}\,(D-1)/D$ returns exactly 1/2 at $D = 4$ with no fitted quantity, while hedging that this was not a proof. **The hedge was right and the form is now withdrawn.** The framework's own D-dependence is derivable, and it is a different function.

Three ingredients, each computed rather than asserted:

1. **The D-dimensional Friedmann coefficient.** Building the FRW Einstein tensor in $D = 4, \ldots, 7$ gives $G_{tt} = 3, 6, 10, 15\, H^2$, hence $H^2 = 16\pi G_D \rho/((D-1)(D-2))$ — the standard $8\pi G\rho/3$ at $D = 4$. A control confirms the extractor rejects all three prespecified decoy coefficients. **So the target's $\sqrt{6}$ is Friedmann, not numerology.**
2. $T_{dS} = H/2\pi$ **in every dimension.** The static-patch function is $f(r) = 1 - H^2 r^2$ identically once $H^2 = 2\Lambda/((D-1)(D-2))$, so the surface gravity is H with no D in it. Control: for Schwarzschild–de Sitter the surface gravity *is* D-dependent, so this is a property of pure de Sitter and not of the method.
3. **The memory factor is a worldline scalar.** The gap depends only on $\cosh(w - w')$ — no spatial index — so neither the 2/3 nor the kernel shape can carry D; and $a_0 = \frac{2}{3} c/M_1$ together with $a_0 = cH_\Lambda/Z$ forces $\xi = 2Z/3$ uniquely.

Therefore

$$\kappa_D \;=\; \tfrac{1}{2}\sqrt{\frac{6}{(D-1)(D-2)}},$$

which agrees with $\frac{2}{3}(D-1)/D$ at $D = 4$ and nowhere else: the ratio is 1.000, 1.509, 2.029, 2.556, 4.157 at $D = 4, 5, 6, 7, 10$. And the single point cannot select a form — **five** prespecified functions pass through exactly 1/2 at $D = 4$: $2/D$, $\frac{2}{3}(D-1)/D$, $3/(2(D-1))$, the constant $\frac{1}{2}$, and the derived one. The $D = 4$ *value* stands; the *form* does not.

7.4 The residue

With $\sqrt{6}$ traced to Friedmann (§7.3) and $\sqrt{\pi}$ to the odd-dimensional momentum measure (§7.2), the unexplained part of the target is a pure rational:

$$\xi^2 = 2\pi\left(\tfrac{4}{3}\right)^3 \quad\Longleftrightarrow\quad M_1 = \tfrac{4}{3}\,t_\Lambda, \qquad \tfrac{4}{3} = 2\times\tfrac{2}{3},$$

and the 2/3 is the derived memory-force renormalisation. **The entire residue is one factor of 2** — which is the same factor already banked as $Z^2 = 4\,(8\pi/3)$. That part is a re-presentation, not progress, and is labelled as such. The new ground is §§2–6 and 7.2–7.3.

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8. What is not claimed

- **This does not derive $\kappa = \frac{1}{2}$, and it is not a partial derivation.** It closes routes. $\kappa = \frac{1}{2}$ remains **fitted**.
- **No claim is made regarding particle physics, the Standard Model, or unification.** The author publicly withdrew earlier statements of that kind and does not restate them. Nothing here is a theory of everything, and nothing here is evidence for one.

- **Theorem 4 is a statement about this class of moment integrals**, not a general theorem that $\sqrt{\pi}$ cannot arise in physics. It says the *de Sitter correlator's moments* cannot carry a half-integer π-weight at integer p. A different object with a different measure could.
- **The load-bearing structural step in §7.3 is an argument, not a computation.** That the kernel shape and the memory factor carry no D follows from their being proper-time scalars; that is reasoning about the objects, not an evaluation of them.
- **The conclusion of §3 is, in one sense, standard QFT wearing a costume.** Any kernel with $1/s^2$ short-distance behaviour has a logarithmically divergent first moment; that is the same logarithm that makes mass renormalisation necessary. A referee will not find the divergence surprising. What is new is *where it lands* — on the unique moment the action is permitted to use — and the π-parity tie in §5.
- **The interpolating function is not new.** $\nu = \sqrt{1 + 1/y}$ is Eq. (9) of [1].
- **Milgrom's own assessment of the vacuum program** is that an actual inertia-from-vacuum mechanism is still far off. This paper does not change that. It narrows it.

What a referee should attack first

1. **The identification of K with a vacuum autocorrelation.** This is the paper's one physical postulate. If K is instead a *response* kernel, everything changes — but not favourably: for a free field the commutator $(1 + n) - n = 1$ is exactly temperature-independent, so the retarded kernel carries no H and could never have supplied a memory scale at all. The temperature must live in the noise kernel. That is verified as a control, and it closes the obvious alternative.
2. **The conformal coupling.** A minimally coupled massless scalar in de Sitter is infrared-pathological and was not used. Whether the physical kernel is conformally coupled is not established here.
3. **The midpoint rule.** $\theta = (s/c)|a(\tau - s/2)| + O(s^3)$ is exact for hyperbolic motion and third order otherwise. A kernel with support at large s probes that error.

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9. Prior work

[1] M. Milgrom, *The modified dynamics as a vacuum effect*, Phys. Lett. A **253**, 273 (1999) [arXiv:astro-ph/9805346]. **The closest prior art, and closer than expected.** Same physical picture; the route goes through the *temperature*, not through a kernel moment, so it never encounters ζ's pole. The interpolating function used here is his Eq. (9).

[2] M. Milgrom, *Dynamics with a non-standard inertia-acceleration relation*, Ann. Phys. **229**, 384 (1994). Modified-inertia theories are generically time-nonlocal. The memory-kernel framing is his.

[3] C. P. Zimmerman, *A causal variational worldline action for modified inertia: the rapidity gap*, Zenodo, DOI 10.5281/zenodo.21845411 (concept). Supplies the action, the equivalence $\kappa = \frac{1}{2} \Leftrightarrow M_1 = \frac{4}{3}t_\Lambda$, the memory force and the $\frac{2}{3}$.

[4] G. W. Gibbons and S. W. Hawking, *Cosmological event horizons, thermodynamics, and particle creation*, Phys. Rev. D **15**, 2738 (1977).

[5] T. S. Bunch and P. C. W. Davies, Proc. R. Soc. Lond. A **360**, 117 (1978).

[6] H. Narnhofer, I. Peter and W. Thirring, Int. J. Mod. Phys. B **10**, 1507 (1996).

[7] S. Deser and O. Levin, *Accelerated detectors and temperature in (anti-) de Sitter spaces*, Class. Quantum Grav. **14**, L163 (1997); see also T. Jacobson, arXiv:gr-qc/9709048. Source of the $\sqrt{a^2+H^2}$ structure that [1] uses.

The Mellin transform of $\sinh^{-2}$, the KMS condition, $\mathfrak{so}(4) = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, the D_2 root system, Macdonald's volume formula, $\mathrm{Vol}(S^{n-1}) = 2\pi^{n/2}/\Gamma(n/2)$, D-dimensional FRW and Tangherlini's solution (Nuovo Cim. **27**, 636 (1963)) are all classical.

A literature check was performed for this paper. No prior work was found computing the moments of the de Sitter worldline correlator, expressing them as $\Gamma(p+1)\zeta(p)$, or connecting $p = 1$ to an acceleration scale. The absence is unsurprising: the question only exists once one posits $M_1 = c/a_0$, which is a relation of [3].

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10. AI-assistance disclosure

Portions of the analysis, numerical verification and drafting were carried out with the assistance of a large language model (Anthropic Claude). The author directed the work, specified and reviewed every load-bearing calculation, and takes full responsibility including for any errors. No AI system satisfies the criteria for authorship and none is listed as an author. Several intermediate claims produced during the work were found to be incorrect and were withdrawn before this deposit — among them the "hundreds of rationals" figure corrected in §7.2 and the $\frac{2}{3}(D-1)/D$ form withdrawn in §7.3.

11. Reproducibility

Every quantitative claim above is produced by a committed script that exits non-zero if any internal consistency check fails, and each carries negative controls that must trip. The three scripts are included in this deposit:

- `mi_wightman_first_moment_2026.py` — §§2–6, 30/30 checks.
- `mi_lorentz_mode_sum_2026.py` — §7.2, 23/23 checks.
- `mi_kappa_D_dependence_rigidity_2026.py` — §7.3, 23/23 checks.

Both footings are carried on every dimensionful number.